

A Methodology for Predictive Soil Mapping of the Rio Bravo Conservation and Managment Area and Beyond

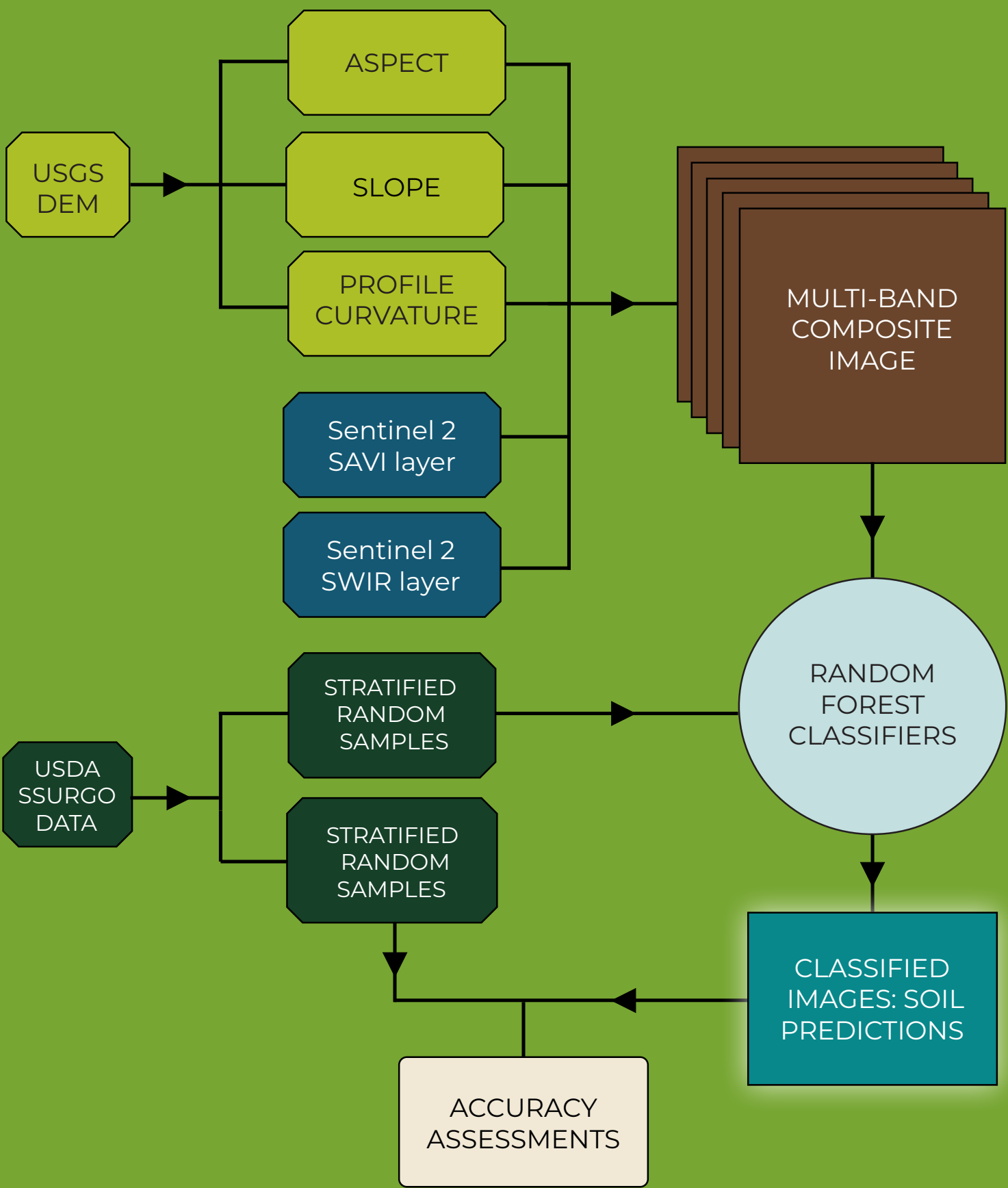
Avery Dyson and Emma Dyson, The University of Texas at Austin

Background and Introduction

The Rio Bravo Conservation and Management Area (RBCMA) of northwest Belize is a vital region for the conservation of natural resources and the preservation of cultural heritage in an area undergoing severe deforestation. Geoarchaeological studies in the region over the last two decades have emphasized the effects of ancient Maya ingenuity on soil formation over the last millennia. However, that research is hindered by the area's remoteness, often making field work—and thus traditional soil mapping—expensive and time-consuming. Still, there remains a need for the ability of a soil map to enhance our understanding of the spatial variability of soils as they relate to climate, vegetation, geology, topography, and time. Here we aim to establish a methodology for the creation of the first digital soil map of this region, which will aid in efforts to conserve biodiversity and to promote future research by improving understandings of soil geographies along this karst landscape.

Barring a current lack of well-distributed soil sample data in the RBCMA, this study tests a methodology using a similarly karst landscape in Texas for which robust soil survey data is already available. From the development of this model digital soil map, we gain valuable lessons as to which data and procedures will be essential to a successful predictive soil mapping effort in the RBCMA and beyond.

METHODOLOGY: DRSNA



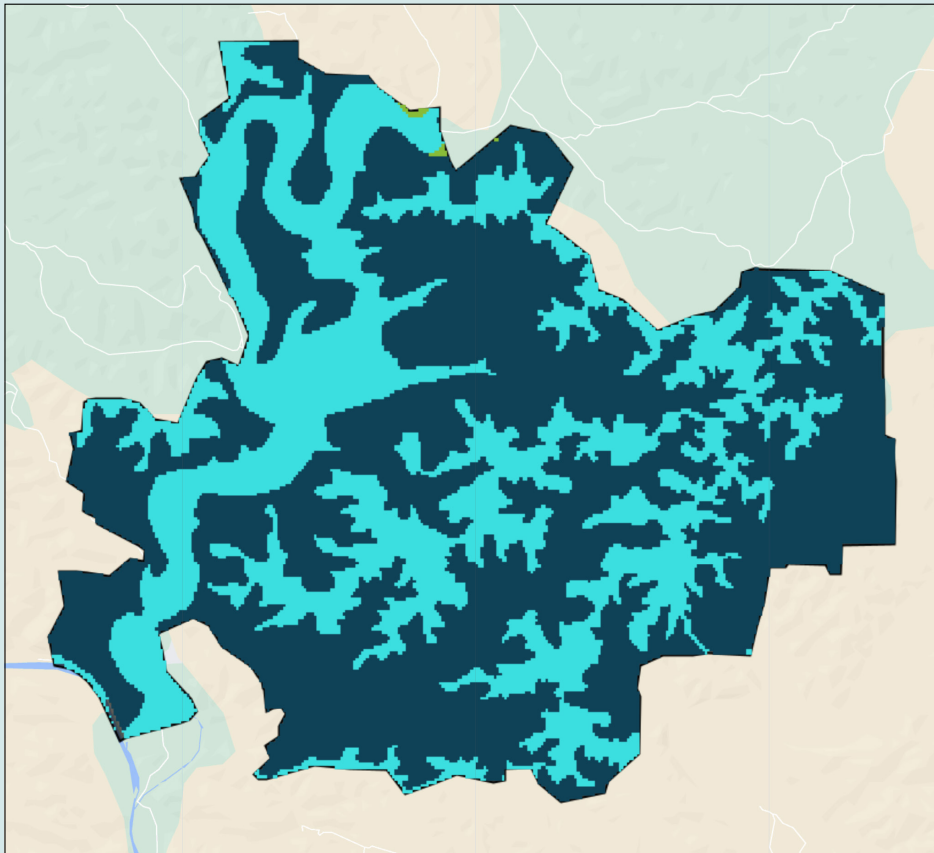
AOI Selection

To establish a DSM methodology appropriate for the soil formation factors acting within the Rio Bravo Conservation and Management Area (RBCMA), we selected the Devils River State Natural Area del Norte (DRSNA) in Texas as our AOI. This area, roughly 19,000 acres, is comparable to the roughly 15,000 acre area within the RBCMA. Although climate and annual rainfall vary between the two regions, both areas are characterized by underlying karst geology and topographical features influenced by prominent river systems. Moreover, the USDA Web Soil Survey (WSS), provided by the National Cooperative Soil Survey, offers robust data on soil characteristics in the Devils River area. It is these data, which can be used for both training and accuracy assessment, that allow us to create and test our predictive soil mapping methods in preparation for their application to the RBCMA and beyond.

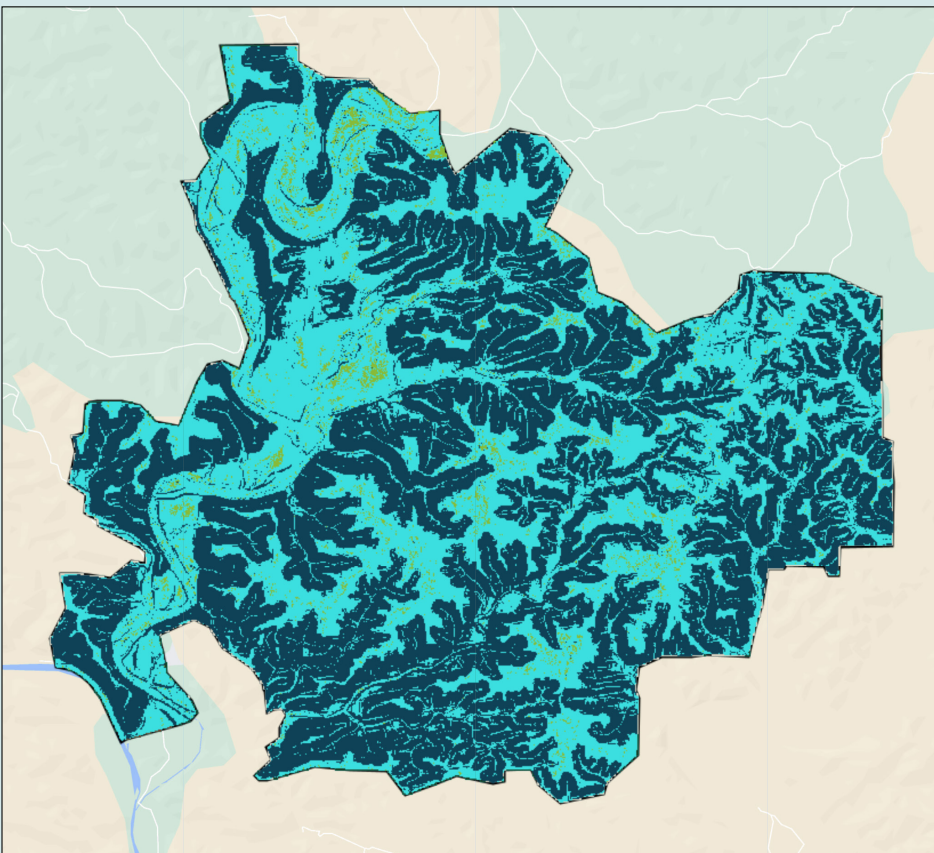
Implementation

The implementation of this workflow was accomplished by first preprocessing topographical soil formation factor images using 1/3 arc second DEM-derived aspect, slope, and profile curvature rasters. We then downloaded NRCS SSURGO data, processing the information using Excel and ArcGIS Pro. Sentinel 2 derived geologic and vegetation soil formation factor images were downloaded and processed using Google Earth Engine. Following preliminary image processing, classification and accuracy assessments were performed in Google Earth Engine, in which we wrote over 700 lines of code in JavaScript. To predict particle size classes, we used 90 training points and a Random Forest classifier with 30 trees, while for subgroup classes, we employed 80 training points and a Random Forest classifier with 40 trees.

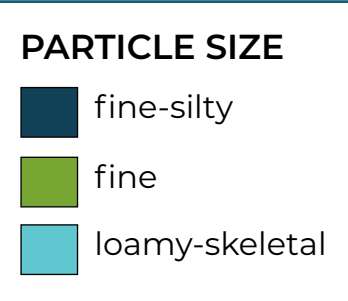
SOIL PARTICLE SIZE CLASSES



SSURGO DATA



PREDICTED



ACCURACY STATISTICS		
PARTICLE SIZE CLASS	USER'S	PRODUCER'S
loamy-skeletal	73.33%	47.83%
fine	30%	100%
fine-silty	90%	77.14%
OVERALL	64.44%	
kappa	0.467	

Devils River State Natural Area del Norte, Texas, USA

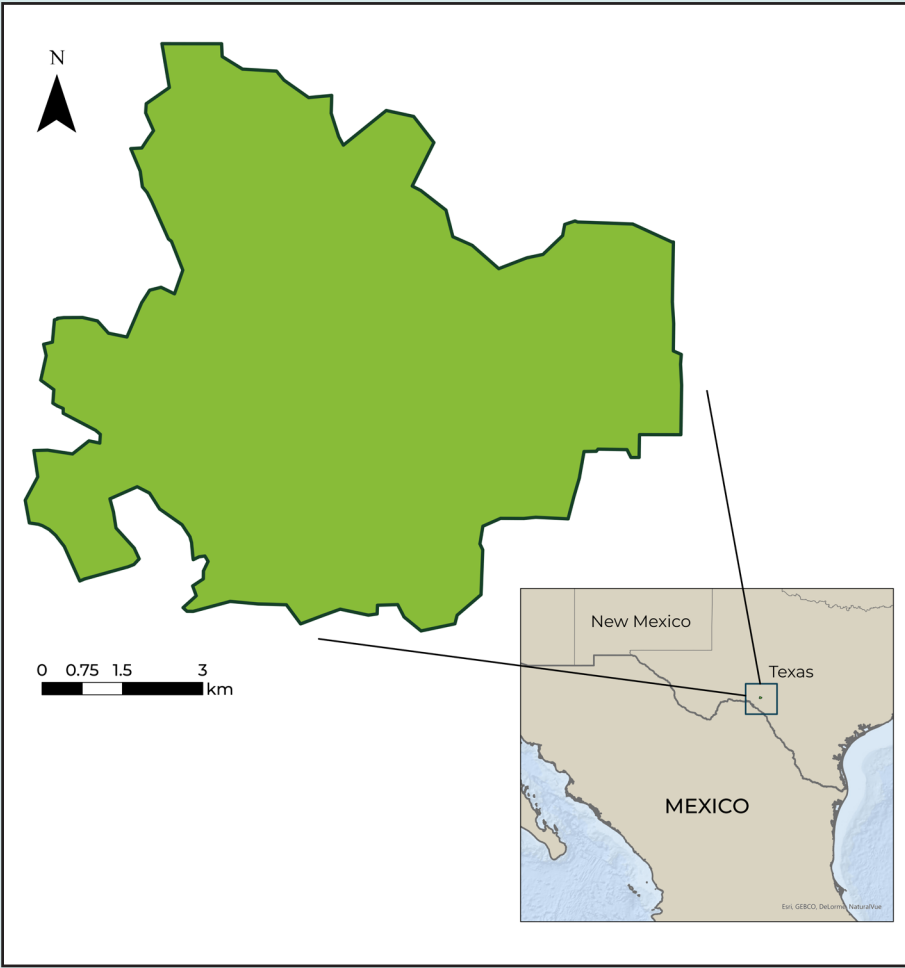
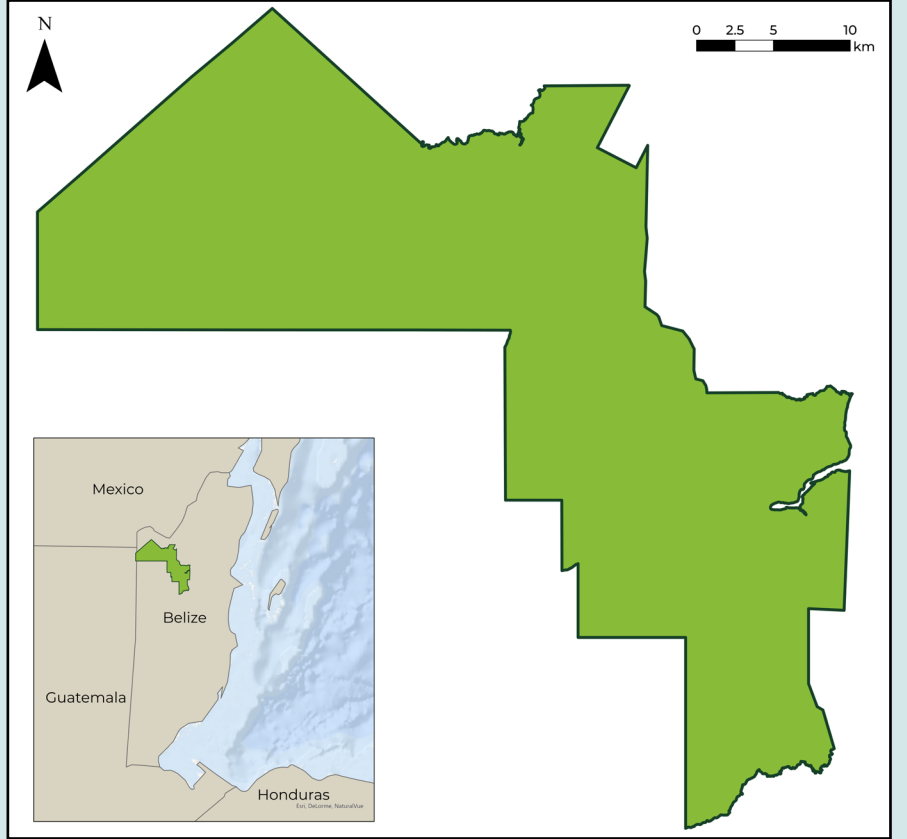


Image source: Texas Parks and Wildlife



Image Source: The Nature Conservancy



Rio Bravo Conservation and Managment Area, Belize

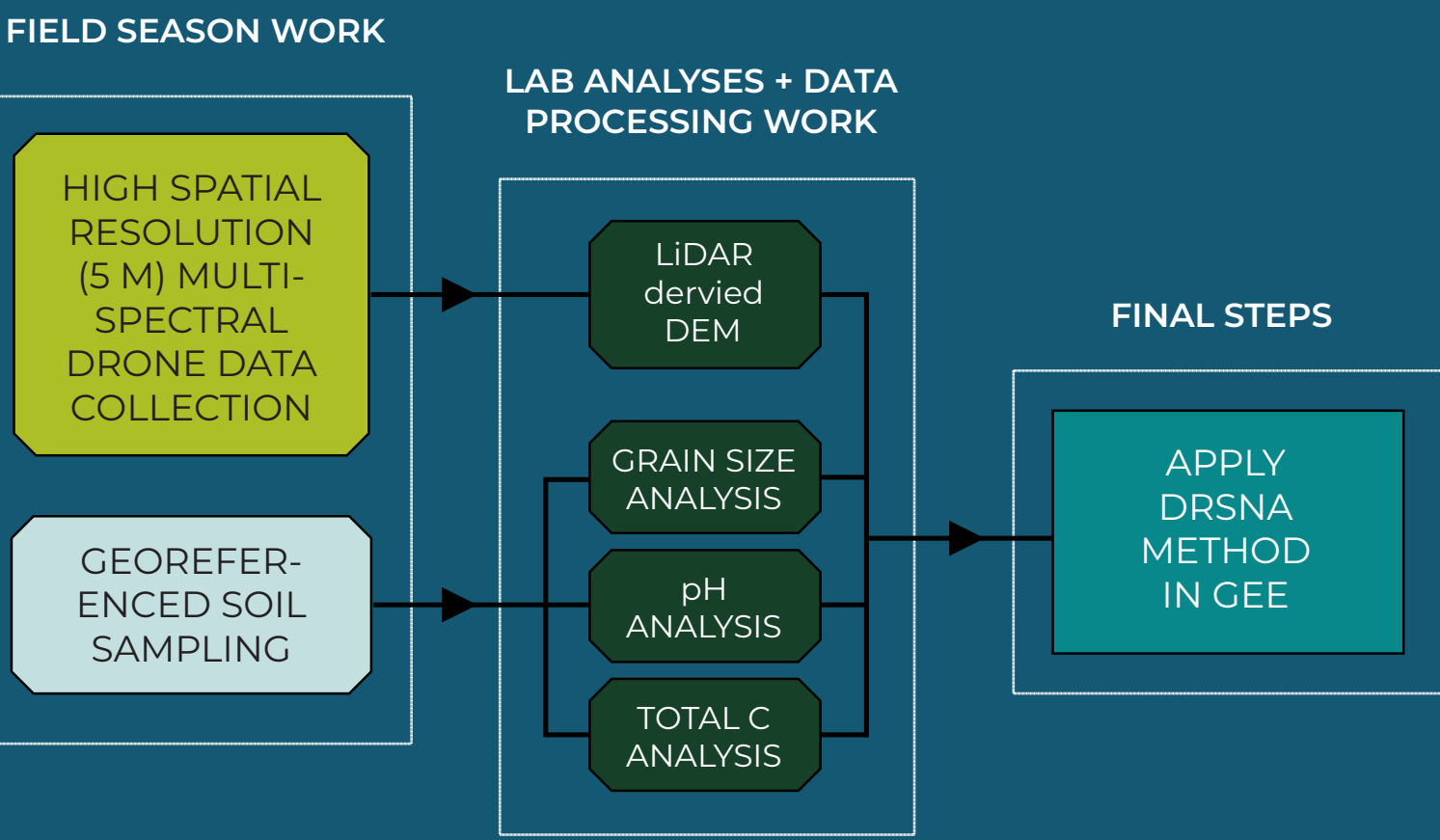
Results and Discussion

The results of our predictive classifications indicate varying degrees of accuracy within particle size class predictions and taxonomic subgroup predictions. The overall accuracy of subgroup classification and particle size classification were 60.83% and 64.44%, respectively, suggesting that our soil classifier method performs slightly better in predicting particle size than taxonomic subgroups. The kappa values for the classifications were 0.489 and 0.467, respectively, indicating moderate agreement between predicted and observed values for both classifications.

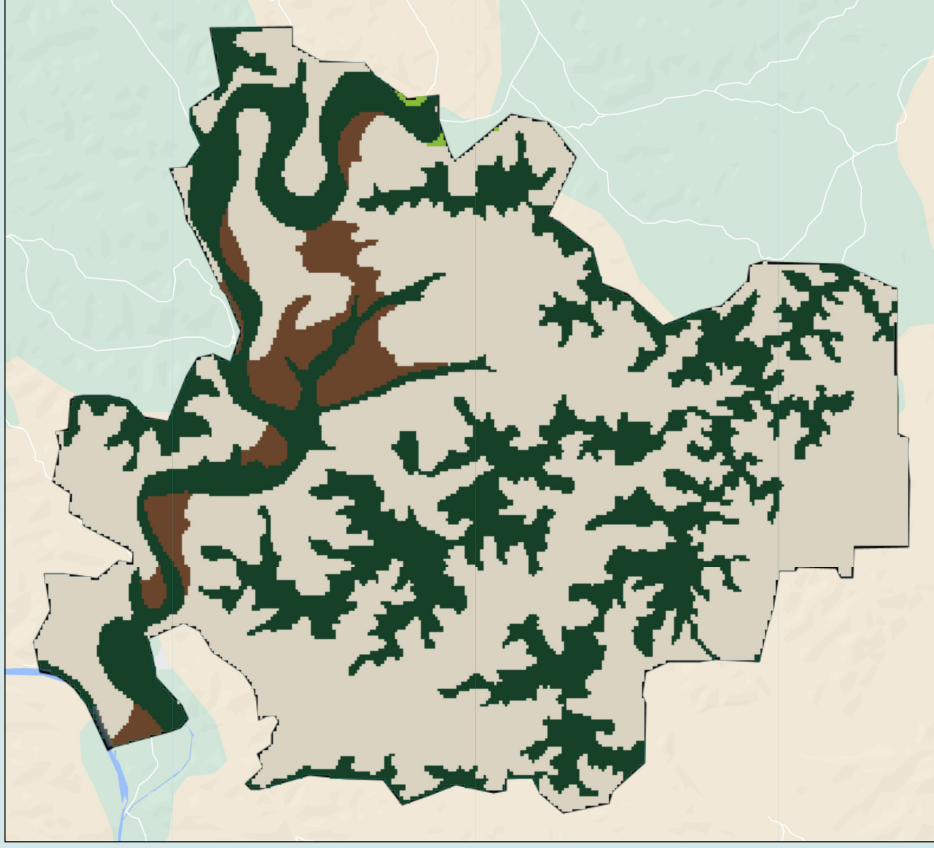
Inaccuracy in these predictive soil maps may be due to several reasons, including the complexity of soil variability and lack of higher resolution imagery. Because soil is a complex and dynamic system, and its properties can vary greatly within short distances, using higher-resolution remote sensing data and more advanced classification algorithms may help us to bridge the gap between current accuracy metrics and a level of excellency (85% accuracy or above). The collection of fine-resolution LiDAR data using a UAV/drones could aid tremendously in this approach.

On the other hand, the fineness of our Sentinel and DEM data may have actually worsened our accuracy as compared to the simplified USDA soil survey we treat as 'ground-truth'. As side-by-side images of our True and Predicted maps demonstrate, USDA Web Soil Survey is highly smoothed and simplified, compared to our predicted images. This is because the Web Soil Survey maps soil characteristics in polygons called 'map units' which are relatively free of fragmentation, perforation, and dissection. While this soil survey data was still the most robust source of training and validation data available, it is important to consider that soil characteristics may actually be varying on a finer scale than demonstrated by the USDA maps. Use of post-processing smoothing functions, or even originally coarser remotely sensed imagery, could have improved statistical accuracy within this project, even as it may have reduced ground-truth accuracy.

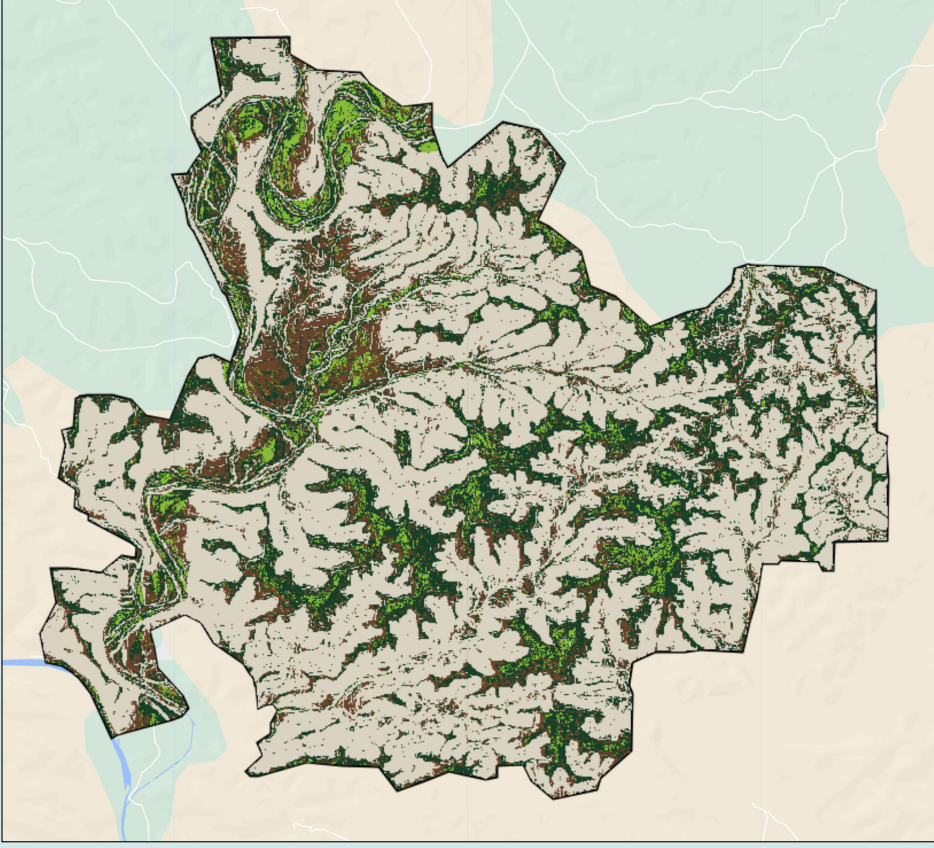
PROSPECTIVE METHODOLOGY: RBCMA



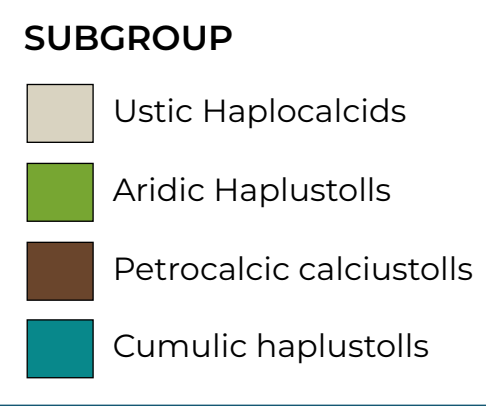
SOIL SUBGROUPS



SSURGO DATA



PREDICTED



ACCURACY STATISTICS		
SUBGROUP CLASS	USER'S	PRODUCER'S
Petrocalcic Calciustolls	46.67%	63.64%
Cumulic Haplustolls	73.33%	46.81%
Aridic Haplustolls	33.33%	100%
Ustic Haplocalcids	93%	68.29%
OVERALL	60.83%	
kappa	0.489	

Authors and Acknowledgments

Avery Dyson, Emma Dyson, Byron Smith, Timothy Beach, Sheryl Luzzadder-Beach. Special thank you to the Beach-Butzer Soils and Geoarchaeology Lab.

Sources: USDA NRCS Soil Survey Geographic (SSURGO) Database Access; Copernicus Sentinel data [2022]; U.S. Geological Survey, 2019; USGS 3D Elevation Program Digital Elevation Model.